



Juan David Guzmán

Structural & Systems Engineer

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Rosario, Argentina

I'm a Civil Engineer specialized in structures, with over 5 years of experience on large-scale industrial projects: oil extraction plants, port terminals and storage silos, for clients such as Cargill, Louis Dreyfus Company, Terminal 6 and COFCO.

SKILLS

STRUCTURAL CALCULATION AND DESIGN

Steel and reinforced concrete structures under CIRSOC, AISC and ACI codes, across every stage: conceptual, basic, detail, fabrication and erection.

SYSTEMS, DATA AND AUTOMATION

Data architecture, operational interfaces and automation, from in-house tools to full web products and AI agent orchestration.

CIRSOC

AISC

ACI

SAP2000

IDEA StatiCa

Tekla Structures

STAAD.Pro

RFEM

AutoCAD 2D/3D

BIM Modeling

AI Automation

Google Apps Script

Python

Firebase

Cloud Firestore

JavaScript

Data architecture

Excel VBA

AppSheet

Looker Studio

Google Drive API

GitHub Pages

Steel Structures Calculation Engineer

Jul 2026 – Present

Flexio · Rosario, Argentina

- Futsal micro-stadium — Jockey Club de Rosario: ~100 t of steel + reinforced concrete infrastructure. Basic and detail engineering, fabrication and erection drawings, direct client relationship.
- Steel portal frame for load testing: calculation and fabrication.
- Introduced calculation-optimization improvements to the technical office.

Structural Engineer

Jan 2024 – Jun 2026

Excel Consulting · Rosario, Argentina

- Design and calculation of steel and concrete structures for oil extraction plants, port terminals and silos.
- Calculation reports, technical specifications and drawings across basic, detail, fabrication and erection engineering. CIRSOC and international standards (Tekla Structures, SAP2000, RFEM).
- Technical audit and validation of contractors' calculation reports and proposals.
- Quantity take-offs and $\pm 30\%$ CAPEX estimates for full industrial plants: Louis Dreyfus Company (5,000 t of steel, 50,000 m³ of concrete) and Terminales y Servicios (2,500 t of steel, 25,000 m³ of concrete).
- Pre-sizing of a 200,000 t storage cell for Cargill Brazil.
- Built in-house engineering tools (Python and Excel VBA) that cut 8-hour tasks down to seconds.

Structural Engineer

Mar 2023 – Dec 2023

AC Ingeniería · Rosario, Argentina

- 34 m tall steel stair tower attached to an existing concrete structure, 3-stage erection, 12 t of steel (SAP2000).
- Calculation, drawings and fabrication control for road signage gantries.
- Calculation, drawings and fabrication supervision of precast concrete elements (400 m²) for Neostar.

Civil Engineer

Aug 2021 – Feb 2023

Deprop SRL · Rosario, Argentina

- Underpinning and expansion of Banco Nación in Firmat: calculation and supervision of underpinning for differential settlement of an in-service institutional building; mixed concrete + steel roof expansion on piles.
- Structural calculation, drawings and supervision for public school classrooms, a kindergarten, and a mixed-structure house on piles.
- Demolition and lintel work for a commercial premises: reinforcement calculation, execution drawings and site follow-up.

Engineering Intern

Mar 2021 – Jun 2021

Ingeniería y Pilotes · Rosario, Argentina

- Site supervision of pile foundations for an industrial building.
- Prepared the qualification file for the Provincial Bidders Registry.

FEATURED PROJECTS

Futsal Micro-Stadium — Jockey Club de Rosario

2026

Roughly 100 tons of steel structure plus reinforced concrete infrastructure for a futsal micro-stadium. Full scope: basic and detail engineering, 3D modeling, fabrication and erection drawings, and direct client relationship throughout the project.

SAP2000 · Tekla Structures · IDEA StatiCa · CIRSOC

BIM Interoperability — SAP2000 · IDEA StatiCa · Tekla Structures

Ongoing

A workflow connecting structural analysis (SAP2000), connection design (IDEA StatiCa) and detail modeling (Tekla Structures) to take a project from calculation to fabrication drawings with minimal friction and rework.

SAP2000 · IDEA StatiCa · Tekla Structures

AI Agent Orchestration — 24/7 Infrastructure

Ongoing

A custom AI agent system deployed on a VPS, running continuously to automate development, technical analysis and management tasks. Modular architecture built for autonomy and operational robustness.

Claude · Automation · 24/7 VPS

EDUCATION & TEACHING

Specialization in Structural Engineering — Universidad Nacional de Rosario (UNR)

2023 – 2024

Civil Engineering — Universidad Nacional de Rosario (UNR)

2015 – 2021

Lecturer — Applied Statics — Universidad Nacional de Rosario (UNR)

2022+

Cambridge English: First (B2) — Cambridge Assessment English

LANGUAGES

Spanish — Native English — B2 (Cambridge)